

Chinese Remainder Theorem

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1 Introduction

Lets have a look at values of x that satisfy the following equations:

$$\begin{aligned}x &\equiv 2 \pmod{7} \\ x &\equiv 1 \pmod{3}\end{aligned}$$

We can think about solutions satisfying each of the individual congruences. To the first one:

$$A = \{\dots, -19, -12, -5, 2, 9, 16, 23, \dots\}$$

And the second one:

$$B = \{\dots, -8, -5, -2, 1, 4, 7, 10, \dots\}$$

So we simply see that all the possible values of x are $x \in A \cup B$.

1.1 Other possible cases

There are also some other cases to think about:

$$\begin{aligned}x &\equiv 1 \pmod{3} \\ x &\equiv 1 \pmod{6}\end{aligned}$$

Or consider this one:

$$\begin{aligned}x &\equiv 1 \pmod{6} \\ x &\equiv 2 \pmod{9}\end{aligned}$$

What happens in each case when we try to find the general solution? These experiments are left as an exercise to the reader.

2 The theorem itself

Theorem 2.1 – Chinese Remainder Theorem Let $a_1, a_2, \dots, a_n$ be integers, and $b_1, b_2, \dots, b_n$ be pairwise coprime integers. Then the system of equations

$$\begin{aligned} x &\equiv a_1 \pmod{b_1} \\ x &\equiv a_2 \pmod{b_2} \\ &\vdots \\ x &\equiv a_n \pmod{b_n} \end{aligned}$$

Has a *unique* solution under $\pmod{b_1 b_2 \dots b_n}$.

Proof of CRT. We split the problem into uniqueness and existence:

Uniqueness: Assume that there are two solutions x, y that satisfy all the congruences. Then we must get that $x - y$ is divisible by $b_i \forall i$. However we have the condition that b_i are pairwise coprime, so $x - y$ must be divisible by $N = b_1 b_2 \dots b_n$, and as a result, $x \equiv y \pmod{N}$, contradiction.

Existence: We will proceed by induction on n .

The case $n = 1$ is trivial, so we start from $n = 2$. Consider the congruences

$$\begin{aligned} x &\equiv a_1 \pmod{b_1} \\ x &\equiv a_2 \pmod{b_2} \end{aligned}$$

By Bezout's identity, we have there exist integers m_1, m_2 such that $m_1 b_1 + m_2 b_2 = 1$. Lets construct $x = a_2 m_1 b_1 + a_1 m_2 b_2$. We will see that this x is the desired x :

$$x = a_2(1 - m_2 b_2) + a_1 m_2 b_2 = a_2 + m_2 b_2(a_1 - a_2) \equiv a_2 \pmod{b_2}$$

Then the other congruence is satisfied similarly.

Assume that there exists a unique x such that n of these congruences hold true simultaneously.

Consider $n + 1$ congruences:

$$\begin{aligned} x &\equiv a_1 \pmod{b_1} \\ x &\equiv a_2 \pmod{b_2} \\ &\vdots \\ x &\equiv a_n \pmod{b_n} \\ x &\equiv a_{n+1} \pmod{b_{n+1}} \end{aligned}$$

We can combine these like so:

$$\begin{aligned} x &\equiv a_{1,2,\dots,n} \pmod{b_{1,2,\dots,n}} \\ x &\equiv a_{n+1} \pmod{b_{n+1}} \end{aligned}$$

And here we have our $n = 2$ base case which we know to work.

□

3 Applications of the theorem

This theorem is extremely useful when we want to construct an x that satisfies many properties.

Problem 3.1 Show that for $c \in \mathbb{Z}$ and a prime p , the congruence $x^x \equiv c \pmod{p}$ has a solution.

The idea is that we are looking at a prime modulus, so FLT probably is going to be useful. Note that $x^{p-1} \equiv 1 \pmod{p}$, so the modulus cycles every $p-1$.

Consider the system of congruences:

$$\begin{aligned}x &\equiv 1 \pmod{p-1} \\x &\equiv c \pmod{p}\end{aligned}$$

Then we know that since $\gcd(p, p-1) = 1$, there is a solution to x by CRT, and hence we have $x^x \equiv x^{x \pmod{p-1}} \equiv x^1 \equiv c \pmod{p}$ and so we are done.

Problem 3.2 What are the last 2 digits of 49^{19} ?

So we are interested in $49^{19} \pmod{100}$. Notice that $100 = 25 \times 4$ and $\gcd(25, 4) = 1$. This means that we only really need to find the x such that

$$\begin{aligned}x &\equiv 49^{19} \pmod{25} \\x &\equiv 49^{19} \pmod{4}\end{aligned}$$

We can compute each of the congruences and turn it into

$$\begin{aligned}x &\equiv -1 \pmod{25} \\x &\equiv 1 \pmod{4}\end{aligned}$$

Which, by CRT, has unique solution $x \equiv ((-1)(4)(19) + (1)(25)(1)) \equiv 49 \pmod{100}$, so we know that the last 2 digits of x are 49.